AC Optimal Power Flow in Rectangular Coordinates with Augmented Lagrangian Formulation

1 Rectangular-Coordinate ACOPF Model

1.1 Sets and Parameters

- $\mathcal{N}$: set of buses, indexed by i, j.
- $\mathcal{G} \subseteq \mathcal{N}$: set of generator buses.
- $\mathcal{L}$: set of transmission lines (i, j).
- P_{Di}, Q_{Di}: active and reactive load at bus i.
- $Y_{ij} = G_{ij} + jB_{ij}$: (i, j) entry of the bus admittance matrix. Here G_{ij} and B_{ij} are the real and imaginary parts, respectively.
- For each line $(i, j) \in \mathcal{L}$, $y_{ij} = g_{ij} + jb_{ij}$ is the series admittance of the branch.
- $V_i^{\min}, V_i^{\max}$: lower and upper limits on voltage magnitude at bus i.
- $P_{Gi}^{\min}, P_{Gi}^{\max}$: active power limits of generator $i \in \mathcal{G}$.
- $Q_{Gi}^{\min}, Q_{Gi}^{\max}$: reactive power limits of generator $i \in \mathcal{G}$.
- $S_{ij}^{\text{total, sq}}, S_{ij,\max}^{\text{total, sq}}$: Total power squared and its limit of branch $(i, j) \in \mathcal{L}$.
- $C_i(P_{Gi})$: generation cost function at generator $i \in \mathcal{G}$, typically a convex quadratic or piecewise-linear function of P_{Gi}.

1.2 Decision Variables

The decision (optimization) variables of the rectangular-coordinate ACOPF are

$$
\begin{aligned}
&P_{Gi},\ Q_{Gi}, && \forall i \in \mathcal{G}, \\
&V_i^R,\ V_i^I,\ \mathfrak{V}_i && \forall i \in \mathcal{N}, \\
&P_{ij},\ Q_{ij},\ S_{ij}^{\text{total, sq}} && \forall (i, j) \in \mathcal{L},
\end{aligned}
$$

where

$$V_i = V_i^R + jV_i^I$$

is the complex bus voltage at bus i written in rectangular coordinates, and $P_{ij} + jQ_{ij}$ is the complex power flow from bus i to bus j.

1.3 Explicit Rectangular-Coordinate ACOPF Formulation

The ACOPF in rectangular coordinates can be written as:

$$\min_{\substack{P_G,Q_G,\\ V^R,V^I,\\ P_{ij},Q_{ij}}} \quad \sum_{i\in\mathcal{G}} C_i(P_{Gi}) \tag{1}$$

$$\text{s.t.} \quad P_{Gi} - P_{Di} = V_i^R \sum_{j\in\mathcal{N}} \big(G_{ij}V_j^R - B_{ij}V_j^I\big) + V_i^I \sum_{j\in\mathcal{N}} \big(G_{ij}V_j^I + B_{ij}V_j^R\big), \qquad \forall i \in \mathcal{N}, \tag{2}$$

$$Q_{Gi} - Q_{Di} = V_i^I \sum_{j\in\mathcal{N}} \big(G_{ij}V_j^R - B_{ij}V_j^I\big) - V_i^R \sum_{j\in\mathcal{N}} \big(G_{ij}V_j^I + B_{ij}V_j^R\big), \qquad \forall i \in \mathcal{N}, \tag{3}$$

$$P_{ij} = V_i^R\big[g_{ij}(V_i^R - V_j^R) - b_{ij}(V_i^I - V_j^I)\big] + V_i^I\big[g_{ij}(V_i^I - V_j^I) + b_{ij}(V_i^R - V_j^R)\big], \quad \forall (i,j) \in \mathcal{L}, \tag{4}$$

$$Q_{ij} = V_i^I\big[g_{ij}(V_i^R - V_j^R) - b_{ij}(V_i^I - V_j^I)\big] - V_i^R\big[g_{ij}(V_i^I - V_j^I) + b_{ij}(V_i^R - V_j^R)\big], \quad \forall (i,j) \in \mathcal{L}, \tag{5}$$

$$\mathfrak{V}_i = (V_i^R)^2 + (V_i^I)^2, \qquad \forall i \in \mathcal{N}, \tag{6}$$

$$S_{ij}^{\text{total, sq}} = P_{ij}^2 + Q_{ij}^2, \qquad \forall (i,j) \in \mathcal{L}. \tag{7}$$

$$V_{\text{ref}}^I = 0, \tag{8}$$

$$P_{Gi}^{\min} \le P_{Gi} \le P_{Gi}^{\max}, \qquad \forall i \in \mathcal{G}, \tag{9}$$

$$Q_{Gi}^{\min} \le Q_{Gi} \le Q_{Gi}^{\max}, \qquad \forall i \in \mathcal{G}, \tag{10}$$

$$\mathfrak{V}_i^{\min} \le \mathfrak{V}_i \le \mathfrak{V}_i^{\max}, \qquad \forall i \in \mathcal{N}, \tag{11}$$

$$0 \le S_{ij}^{\text{total, sq}} \le S_{ij,\max}^{\text{total,sq}}, \qquad \forall (i,j) \in \mathcal{L}. \tag{12}$$

The model (1)–(7)

is the ACOPF model in a rectangular coordinate system where all equations are explicitly expanded,

where all decision variables are

$$\{P_{Gi}, Q_{Gi}\}_{i\in\mathcal{G}}, \quad \{V_i^R, V_i^I, \mathfrak{V}_i\}_{i\in\mathcal{N}}, \quad \{P_{ij}, Q_{ij}, S_{ij}^{\text{total, sq}}\}_{(i,j)\in\mathcal{L}}.$$

Now, In order to reduce the order of constraints of the problem. We will do such substitution

$$\begin{bmatrix} V_i^{\Re} \\ V_i^{\Im} \\ V_j^{\Re} \\ V_j^{\Im} \end{bmatrix} \cdot \begin{bmatrix} V_i^{\Re} V_i^{\Im} V_j^{\Re} V_j^{\Im} \end{bmatrix} = \begin{bmatrix} (V_i^{\Re})^2 & V_i^{\Re}V_i^{\Im} & V_i^{\Re}V_j^{\Re} & V_i^{\Re}V_j^{\Im} \\ V_i^{\Im}V_i^{\Re} & (V_i^{\Im})^2 & V_i^{\Im}V_j^{\Re} & V_i^{\Im}V_j^{\Im} \\ V_j^{\Re}V_i^{\Re} & V_j^{\Re}V_i^{\Im} & (V_j^{\Re})^2 & V_j^{\Re}V_j^{\Im} \\ V_j^{\Im}V_i^{\Re} & V_j^{\Im}V_i^{\Im} & V_j^{\Im}V_j^{\Re} & (V_j^{\Im})^2 \end{bmatrix} = \begin{bmatrix} \mathfrak{V}_{i,i}^{\Re,\Re} & \mathfrak{V}_{i,i}^{\Re,\Im} & \mathfrak{V}_{i,j}^{\Re,\Re} & \mathfrak{V}_{i,j}^{\Re,\Im} \\ \mathfrak{V}_{i,i}^{\Im,\Re} & \mathfrak{V}_{i,i}^{\Im,\Im} & \mathfrak{V}_{i,j}^{\Im,\Re} & \mathfrak{V}_{i,j}^{\Im,\Im} \\ \mathfrak{V}_{j,i}^{\Re,\Re} & \mathfrak{V}_{j,i}^{\Re,\Im} & \mathfrak{V}_{j,j}^{\Re,\Re} & \mathfrak{V}_{j,j}^{\Re,\Im} \\ \mathfrak{V}_{j,i}^{\Im,\Re} & \mathfrak{V}_{j,i}^{\Im,\Im} & \mathfrak{V}_{j,j}^{\Im,\Re} & \mathfrak{V}_{j,j}^{\Im,\Im} \end{bmatrix}. \tag{13}$$

Simplify:

$$\begin{bmatrix} \mathfrak{V}_{i,i}^{\Re,\Re} & \mathfrak{V}_{i,i}^{\Re,\Im} & \mathfrak{V}_{i,j}^{\Re,\Re} & \mathfrak{V}_{i,j}^{\Re,\Im} \\ \mathfrak{V}_{i,i}^{\Im,\Re} & \mathfrak{V}_{i,i}^{\Im,\Im} & \mathfrak{V}_{i,j}^{\Im,\Re} & \mathfrak{V}_{i,j}^{\Im,\Im} \\ \mathfrak{V}_{j,i}^{\Re,\Re} & \mathfrak{V}_{j,i}^{\Re,\Im} & \mathfrak{V}_{j,j}^{\Re,\Re} & \mathfrak{V}_{j,j}^{\Re,\Im} \\ \mathfrak{V}_{j,i}^{\Im,\Re} & \mathfrak{V}_{j,i}^{\Im,\Im} & \mathfrak{V}_{j,j}^{\Im,\Re} & \mathfrak{V}_{j,j}^{\Im,\Im} \end{bmatrix} = \begin{bmatrix} \mathfrak{V}_{i,i}^{\Re,\Re} & \mathfrak{V}_{i,i}^{\Re,\Im} & \mathfrak{V}_{i,j}^{\Re,\Re} & \mathfrak{V}_{i,j}^{\Re,\Im} \\ \mathrm{O} & \mathfrak{V}_{i,i}^{\Im,\Im} & \mathfrak{V}_{i,j}^{\Im,\Re} & \mathfrak{V}_{i,j}^{\Im,\Im} \\ \mathrm{O} & \mathrm{O} & \mathfrak{V}_{j,j}^{\Re,\Re} & \mathfrak{V}_{j,j}^{\Re,\Im} \\ \mathrm{O} & \mathrm{O} & \mathrm{O} & \mathfrak{V}_{j,j}^{\Im,\Im} \end{bmatrix} \tag{14}$$

Then the ACOPF becomes

$$\min_{\substack{P_G,Q_G,\\ V^R,V^I,\\ P_{ij},Q_{ij},\\ \mathfrak{V}}} \quad \sum_{i\in\mathcal{G}} C_i(P_{Gi}) \tag{15}$$

$$\text{s.t.} \quad P_{Gi} - P_{Di} = \sum_{j\in\mathcal{N}} \Big(G_{ij}\mathfrak{V}_{i,j}^{\Re,\Re} - B_{ij}\mathfrak{V}_{i,j}^{\Re,\Im} + G_{ij}\mathfrak{V}_{i,j}^{\Im,\Im} + B_{ij}\mathfrak{V}_{i,j}^{\Im,\Re} \Big), \qquad \forall i \in \mathcal{N}, \tag{16}$$

$$Q_{Gi} - Q_{Di} = \sum_{j\in\mathcal{N}} \Big(G_{ij}\mathfrak{V}_{i,j}^{\Im,\Re} - B_{ij}\mathfrak{V}_{i,j}^{\Im,\Im} - G_{ij}\mathfrak{V}_{i,j}^{\Re,\Im} - B_{ij}\mathfrak{V}_{i,j}^{\Re,\Re} \Big), \qquad \forall i \in \mathcal{N}, \tag{17}$$

$$P_{ij} = g_{ij}\big(\mathfrak{V}_{i,i}^{\Re,\Re} + \mathfrak{V}_{i,i}^{\Im,\Im}\big) - g_{ij}\big(\mathfrak{V}_{i,j}^{\Re,\Re} + \mathfrak{V}_{i,j}^{\Im,\Im}\big) - b_{ij}\mathfrak{V}_{i,i}^{\Re,\Im} + b_{ij}\mathfrak{V}_{i,j}^{\Re,\Im} + b_{ij}\mathfrak{V}_{i,i}^{\Im,\Re} - b_{ij}\mathfrak{V}_{i,j}^{\Im,\Re}, \quad \forall (i,j) \in \mathcal{L}, \tag{18}$$

$$Q_{ij} = g_{ij}\mathfrak{V}_{i,i}^{\Im,\Re} - g_{ij}\mathfrak{V}_{i,j}^{\Im,\Re} - b_{ij}\mathfrak{V}_{i,i}^{\Im,\Im} + b_{ij}\mathfrak{V}_{i,j}^{\Im,\Im} - g_{ij}\mathfrak{V}_{i,i}^{\Re,\Im} + g_{ij}\mathfrak{V}_{i,j}^{\Re,\Im} - b_{ij}\mathfrak{V}_{i,i}^{\Re,\Re} + b_{ij}\mathfrak{V}_{i,j}^{\Re,\Re}, \quad \forall (i,j) \in \mathcal{L}, \tag{19}$$

$$\mathfrak{V}_i = \mathfrak{V}_{i,i}^{\Re,\Re} + \mathfrak{V}_{i,i}^{\Im,\Im}, \qquad \forall i \in \mathcal{N}, \tag{20}$$

$$S_{ij}^{\text{total,sq}} = P_{ij}^2 + Q_{ij}^2, \qquad \forall (i,j) \in \mathcal{L}, \tag{21}$$

$$\mathfrak{V}_{\text{ref}}^{I} = 0, \tag{22}$$

$$P_{Gi}^{\min} \le P_{Gi} \le P_{Gi}^{\max}, \qquad \forall i \in \mathcal{G}, \tag{23}$$

$$Q_{Gi}^{\min} \le Q_{Gi} \le Q_{Gi}^{\max}, \qquad \forall i \in \mathcal{G}, \tag{24}$$

$$\mathfrak{V}_i^{\min} \le \mathfrak{V}_i \le \mathfrak{V}_i^{\max}, \qquad \forall i \in \mathcal{N}, \tag{25}$$

$$0 \le S_{ij}^{\text{total,sq}} \le S_{ij,\max}^{\text{total,sq}}, \qquad \forall (i,j) \in \mathcal{L}. \tag{26}$$

Simplify the equation

$$
\begin{aligned}
\min_{\substack{P_G, Q_G,\\ \mathfrak{V}, P_{ij}, Q_{ij},}} \quad & \sum_{i\in\mathcal{G}} C_i(P_{Gi}) && && (27)\\
\text{s.t.} \quad & P_{Gi} - P_{Di} = \sum_{j\in\mathcal{N}} \Big(G_{ij}\mathfrak{V}^{\Re,\Re}_{i,j} - B_{ij}\mathfrak{V}^{\Re,\Im}_{i,j} + G_{ij}\mathfrak{V}^{\Im,\Im}_{i,j} + B_{ij}\mathfrak{V}^{\Im,\Re}_{i,j} \Big), && \forall i\in\mathcal{N}, && (28)\\
& Q_{Gi} - Q_{Di} = \sum_{j\in\mathcal{N}} \Big(G_{ij}\mathfrak{V}^{\Im,\Re}_{i,j} - B_{ij}\mathfrak{V}^{\Im,\Im}_{i,j} - G_{ij}\mathfrak{V}^{\Re,\Im}_{i,j} - B_{ij}\mathfrak{V}^{\Re,\Re}_{i,j} \Big), && \forall i\in\mathcal{N}, && (29)\\
& P_{ij} = g_{ij}\Big(\mathfrak{V}^{\Re,\Re}_{i,i} + \mathfrak{V}^{\Im,\Im}_{i,i} - \mathfrak{V}^{\Re,\Re}_{i,j} - \mathfrak{V}^{\Im,\Im}_{i,j} \Big) + b_{ij}\Big(\mathfrak{V}^{\Re,\Im}_{i,j} - \mathfrak{V}^{\Im,\Re}_{i,j} \Big), && \forall (i,j)\in\mathcal{L}, && (30)\\
& Q_{ij} = b_{ij}\Big(\mathfrak{V}^{\Re,\Re}_{i,j} + \mathfrak{V}^{\Im,\Im}_{i,j} - \mathfrak{V}^{\Re,\Re}_{i,i} - \mathfrak{V}^{\Im,\Im}_{i,i} \Big) + g_{ij}\Big(\mathfrak{V}^{\Re,\Im}_{i,j} - \mathfrak{V}^{\Im,\Re}_{i,j} \Big), && \forall (i,j)\in\mathcal{L}, && (31)\\
& \mathfrak{V}_i = \mathfrak{V}^{\Re,\Re}_{i,i} + \mathfrak{V}^{\Im,\Im}_{i,i}, && \forall i\in\mathcal{N}, && (32)\\
& S^{\text{total,sq}}_{ij} = P_{ij}^2 + Q_{ij}^2, && \forall (i,j)\in\mathcal{L}, && (33)\\
& \mathfrak{V}^{\Im,\Im}_{\text{ref},j} = \mathfrak{V}^{\Im,\Re}_{\text{ref},j} = 0 && \forall (\text{ref},j)\in\mathcal{L}, && (34)\\
& P^{\min}_{Gi} \le P_{Gi} \le P^{\max}_{Gi}, && \forall i\in\mathcal{G}, && (35)\\
& Q^{\min}_{Gi} \le Q_{Gi} \le Q^{\max}_{Gi}, && \forall i\in\mathcal{G}, && (36)\\
& \mathfrak{V}^{\min}_{i} \le \mathfrak{V}_{i} \le \mathfrak{V}^{\max}_{i}, && \forall i\in\mathcal{N}, && (37)\\
& 0 \le S^{\text{total,sq}}_{ij} \le S^{\text{total,sq}}_{ij,\max}, && \forall (i,j)\in\mathcal{L}. && (38)
\end{aligned}
$$

2 Augmented Lagrangian Reformulation (Lifted Rectangular Form)

Let

$$\mathbf{x} = \big(P_G,\, Q_G,\, \mathfrak{V},\, P_{ij},\, Q_{ij},\, S^{\text{total, sq}}\big)$$

collect all primal variables, where

$$\mathfrak{V}^{\text{lift}} := \big(\mathfrak{V}^{\alpha,\beta}_{a,b}\big)_{\substack{a,b\in\mathcal{N}\\ \alpha,\beta\in\{\Re,\Im\}}}, \qquad \mathfrak{V} = (\mathfrak{V}_i)_{i\in\mathcal{N}}, \qquad S^{\text{total, sq}} = \big(S^{\text{total, sq}}_{ij}\big)_{(i,j)\in\mathcal{L}}.$$

We define the lifted variables by

$$\mathfrak{V}^{\alpha,\beta}_{a,b} := V^{\alpha}_{a} V^{\beta}_{b}, \qquad \forall a,b\in\mathcal{L},\ \forall \alpha,\beta\in\{\Re,\Im\}. \tag{39}$$

We now treat *all* equality constraints in the model, including bus power balance, branch-flow definitions, lifted-variable definitions, auxiliary-variable definitions, and the reference-bus condition, as augmented constraints.

Bus power balance constraints. For each bus $i\in\mathcal{N}$,

$$c^P_i(\mathbf{x}) := P_{Gi} - P_{Di} - \sum_{j\in\mathcal{N}} \Big(G_{ij}\mathfrak{V}^{\Re,\Re}_{i,j} - B_{ij}\mathfrak{V}^{\Re,\Im}_{i,j} + G_{ij}\mathfrak{V}^{\Im,\Im}_{i,j} + B_{ij}\mathfrak{V}^{\Im,\Re}_{i,j} \Big) = 0, \tag{40}$$

$$c^Q_i(\mathbf{x}) := Q_{Gi} - Q_{Di} - \sum_{j\in\mathcal{N}} \Big(G_{ij}\mathfrak{V}^{\Im,\Re}_{i,j} - B_{ij}\mathfrak{V}^{\Im,\Im}_{i,j} - G_{ij}\mathfrak{V}^{\Re,\Im}_{i,j} - B_{ij}\mathfrak{V}^{\Re,\Re}_{i,j} \Big) = 0. \tag{41}$$

Branch power-flow definition constraints. For each line $(i, j) \in \mathcal{L}$,

$$c_{ij}^{P,\text{flow}}(\mathbf{x}) \coloneqq P_{ij} - \left[g_{ij} \left(\mathfrak{V}_{i,i}^{\Re,\Re} + \mathfrak{V}_{i,i}^{\Im,\Im} - \mathfrak{V}_{i,j}^{\Re,\Re} - \mathfrak{V}_{i,j}^{\Im,\Im} \right) + b_{ij} \left(\mathfrak{V}_{i,j}^{\Re,\Im} - \mathfrak{V}_{i,j}^{\Im,\Re} \right) \right] = 0, \tag{42}$$

$$c_{ij}^{Q,\text{flow}}(\mathbf{x}) \coloneqq Q_{ij} - \left[b_{ij} \left(\mathfrak{V}_{i,j}^{\Re,\Re} + \mathfrak{V}_{i,j}^{\Im,\Im} - \mathfrak{V}_{i,i}^{\Re,\Re} - \mathfrak{V}_{i,i}^{\Im,\Im} \right) + g_{ij} \left(\mathfrak{V}_{i,j}^{\Re,\Im} - \mathfrak{V}_{i,j}^{\Im,\Re} \right) \right] = 0. \tag{43}$$

Lifted-variable definition constraints. For all required indices $a, b \in \mathcal{N}$ and $\alpha, \beta \in \{\Re, \Im\}$,

$$c_{a,b}^{\alpha,\beta}(\mathbf{x}) \coloneqq \mathfrak{V}_{a,b}^{\alpha,\beta} - V_a^\alpha V_b^\beta = 0. \tag{44}$$

Voltage-magnitude auxiliary-variable constraints. For each bus $i \in \mathcal{N}$,

$$c_i^{\mathfrak{V}}(\mathbf{x}) \coloneqq \mathfrak{V}_i - \left(\mathfrak{V}_{i,i}^{\Re,\Re} + \mathfrak{V}_{i,i}^{\Im,\Im} \right) = 0. \tag{45}$$

Branch total power (squared) auxiliary-variable constraints. For each line $(i, j) \in \mathcal{L}$,

$$c_{ij}^{S}(\mathbf{x}) \coloneqq S_{ij}^{\text{total, sq}} - \left(P_{ij}^2 + Q_{ij}^2 \right) = 0. \tag{46}$$

However, This constraints is quadratic. We need to linearize it to make it fit to QHD. We will use such linearization to do things. Therefore, the linearized auxiliary-variable constraint is written as

$$c_{ij}^{S,\text{lin}}(\mathbf{x}; \mathbf{x}^{(k)}) \coloneqq S_{ij}^{\text{total,sq}} - \left(2P_{ij}^{(k)} P_{ij} + 2Q_{ij}^{(k)} Q_{ij} - \left(P_{ij}^{(k)}\right)^2 - \left(Q_{ij}^{(k)}\right)^2 \right) = 0. \tag{47}$$

Reference-bus constraint. For the reference bus,

$$c^{\text{ref}}(\mathbf{x}) \coloneqq \mathfrak{V}_{\text{ref},j}^{\Re\Im} = \mathfrak{V}_{j,\text{ref}}^{\Im\Re} = \mathfrak{V}_{\text{ref,ref}}^{\Im\Im} = 0. \tag{48}$$

Collecting all these constraints, we define

$$h(\mathbf{x}) = \begin{bmatrix} \{c_i^P(\mathbf{x})\}_{i \in \mathcal{N}} \\ \{c_i^Q(\mathbf{x})\}_{i \in \mathcal{N}} \\ \{c_{ij}^{P,\text{flow}}(\mathbf{x})\}_{(i,j) \in \mathcal{L}} \\ \{c_{ij}^{Q,\text{flow}}(\mathbf{x})\}_{(i,j) \in \mathcal{L}} \\ \{c_{a,b}^{\alpha,\beta}(\mathbf{x})\}_{\substack{a,b \in \mathcal{L} \\ \alpha,\beta \in \{\Re,\Im\}}} \\ \{c_i^{\mathfrak{V}}(\mathbf{x})\}_{i \in \mathcal{N}} \\ \{c_{ij}^S(\mathbf{x})\}_{(i,j) \in \mathcal{L}} \\ c^{\text{ref}}(\mathbf{x}) \end{bmatrix}. \tag{49}$$

Let the corresponding Lagrange multipliers be λ, stacked in the same order as the components of $h(\mathbf{x})$.

The generation cost is

$$f(\mathbf{x}) = \sum_{i \in \mathcal{G}} C_i(P_{Gi}). \tag{50}$$

2.1 Augmented Lagrangian Function (Fully Augmented)

With all equality constraints included in $h(\mathbf{x})$, the augmented Lagrangian function is

$$\mathcal{L}_\rho(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda^\top h(\mathbf{x}) + \frac{\rho}{2} \|h(\mathbf{x})\|_2^2 + \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}_{\text{prev}}\|_2^2, \tag{51}$$

where $\rho > 0$ is the penalty parameter.

3 Box-Constrained Feasible Set and AL Iterations

After moving all equality constraints into the augmented Lagrangian, the remaining constraints are simple bound (box) constraints on the variables. We denote the feasible set by

$$
\begin{aligned}
\mathcal{X} = \Big\{\mathbf{x} : & P_{Gi}^{\min} \le P_{Gi} \le P_{Gi}^{\max}, \quad \forall i \in \mathcal{G}, && (52)\\
& Q_{Gi}^{\min} \le Q_{Gi} \le Q_{Gi}^{\max}, \quad \forall i \in \mathcal{G}, && (53)\\
& \mathfrak{V}_i^{\min} \le \mathfrak{V}_i \le \mathfrak{V}_i^{\max}, \quad \forall i \in \mathcal{N}, && (54)\\
& 0 \le S_{ij}^{\text{total, sq}} \le S_{ij,\max}^{\text{total, sq}}, \quad \forall (i,j) \in \mathcal{L}, && (55)\\
& \underline{V}_i^R \le V_i^R \le \overline{V}_i^R, \quad \underline{V}_i^I \le V_i^I \le \overline{V}_i^I, \quad \forall i \in \mathcal{N}, && (56)\\
& \underline{P}_{ij} \le P_{ij} \le \overline{P}_{ij}, \quad \underline{Q}_{ij} \le Q_{ij} \le \overline{Q}_{ij}, \quad \forall (i,j) \in \mathcal{L}, && (57)\\
& \underline{\mathfrak{V}}_{a,b}^{\alpha,\beta} \le \mathfrak{V}_{a,b}^{\alpha,\beta} \le \overline{\mathfrak{V}}_{a,b}^{\alpha,\beta}, \quad \forall a,b \in \mathcal{N},\ \forall \alpha,\beta \in \{\Re, \Im\}\Big\}. && (58)
\end{aligned}
$$

In compact notation,

$$\mathcal{X} = \{\mathbf{x} : \underline{\mathbf{x}} \le \mathbf{x} \le \overline{\mathbf{x}}\}.$$

The augmented Lagrangian method then solves the sequence of box-constrained subproblems

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x} \in \mathcal{X}} \mathcal{L}_\rho(\mathbf{x}, \lambda^k), \tag{59}$$

$$\lambda^{k+1} = \lambda^k + \rho\, h(\mathbf{x}^{k+1}), \tag{60}$$

where k is the iteration index.